



COMSATS University Islamabad
Attock Campus
Department Of Computer Engineering

Complex Engineering Activity

Course Code:	Course Title:	Session:	Class:
CPE344	Digital Systems Design Lab	Fall 2025	CE-7
Title of Complex Engineering Activity	Design and implement an FPGA-based encryption/decryption unit (e.g., AES, RSA, SHA) for secure communication AES: Advanced Encryption Standard, RSA: Rivest-Shamir-Adleman, SHA: Secure Hash Algorithm Design Requirements: Students will design and implement a cryptographic hardware engine on FPGA using Verilog HDL. The engine will perform encryption and decryption of digital data using widely adopted cryptographic standards such as AES, RSA, or SHA. The project requires students to: • Understand the mathematical and algorithmic foundations of cryptography. • Translate algorithms into optimized hardware designs in Verilog. • Integrate the hardware with input/output modules for secure communication. • Evaluate performance in terms of security, power efficiency, and resource utilization.		



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CEA Requirements

Statement

Preamble:

With the increasing need for **secure communication** in online banking, e-commerce, healthcare, and defense, encryption plays a crucial role in ensuring **data privacy** and **security**. As digital systems engineers, your task is to design and implement a **cryptographic hardware engine** on an FPGA using **Verilog HDL**.

Your design must be able to:

1. Perform **encryption and decryption** using a standard algorithm (AES, RSA, or SHA).
2. Optimise for **speed, area, and power consumption**.
3. Demonstrate real-time secure communication between FPGA and an external device (e.g., PC, microcontroller, or another FPGA).
4. Highlight **societal impact** of secure hardware systems and importance of **energy-efficient designs**.



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EA1	Range of resources: In this project, you will access a diverse range of resources: • People: Collaborate with peers and experts regarding the cryptographic standards • Technologies: Utilise ModelSim/Xilinx ISE/FPGA • Financial: Work within resource constraints—use available software tools. • Materials: Use the books and internet. You must justify the selection and importance of each resource to meet the system objectives.
EA2	Level of Interaction: You will resolve significant technical problems arising from: • Cryptographic hardware engine design in Verilog HDL • Cryptographic hardware engine Test Bench in Verilog HDL • Implement cryptographic hardware engine on FPGA and evaluate with available standard datasets and/or real-time inputs Your work will demonstrate how multiple domains designs interact and how conflicts are addressed.
EA3	Innovation: You are expected to apply innovative engineering solutions: Apply innovative Digital System Design strategies by: • FPGA Resource Usage: Use Spartan-6 FPGA boards (or equivalent) within available resources. • Clock Frequency: Operate at ≥ 25 MHz for functional demonstration. • Key Length: For AES $\rightarrow$ 128 bits; for RSA $\rightarrow$ minimum 1024 bits; for SHA $\rightarrow$ SHA-256. • Interfaces: UART / SPI / I²C communication to demonstrate secure data transfer. • Power Efficiency: Minimise switching activity and use efficient coding practices. Highlight and justify novelty in the algorithm, implementation, or application.



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EA4	Consequences to Society and the Environment: The overall system you develop can impact: • Society: Recognize the societal and ethical implications of secure communication. • Environment: Consider energy efficiency in the architecture. Analyze consequences related to the reliability and ethical use of Cryptographic hardware engine systems in real-world contexts.
EA5	Familiarity: This project extends beyond prior coursework by requiring: • Use of advanced ModelSim tools and/or FPGA based implementation. • Independent research to design application specific architecture of Cryptographic hardware engine. • Application of industry-relevant Digital Systems Design skills. You are expected to justify your approach using principle-based reasoning and document how you adapted new tools and techniques.
	Deliverables 1. Technical Report • Design Documentation: Prepare a comprehensive design document that details your system's architecture. This should include instruction set architecture, target application, and a clear rationale behind key design decisions, Cycle time modelling, and pipelining methods. • System Diagram: Provide a schematic or block diagram that clearly illustrates the flow of your designed system. 2. Simulation • Modelling and Validation: Use ModelSim to demonstrate the functionality of your designed system. The simulation should cover various scenarios, showcasing trade-offs in design (performance, security, energy efficiency). 3. Presentation



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	• Design Overview: Deliver an engaging presentation that explains your engineering decisions, the performance of your designed system, the challenges you encountered, and the innovative solutions you implemented. • Visual Aids: Use visuals such as flowcharts, and simulation waveforms and encrypted/decrypted text etc., to clearly communicate your findings and insights.
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Rubrics for Assessment

COMPLEX ENGINEERING ACTIVITIES ATTRIBUTES	Descriptors for Rubrics Design	EA (Some or all)	Level-1 Needs Work	Level-2 Developing	Level-3 Satisfactory	Level-4 Competent	Level-5 Excellent
Range of resources use of diverse resources (people, money, information, technologies)	Elaborate functions and associations with different resources such as people, money, equipment, materials, information and technologies	EA1	Associate with 1 resource but no elaboration.	Associate with 1 resource with little elaboration.	Associate with 2 resources with significant elaboration.	Associate with 3 resources with significant elaboration.	Associate with more than 3 resources with very significant elaboration.
	Justify the involvement of these resources in fulfilling the requirements of a successful design project.		Justify on 1 resource only.	Justify on 1 resource with acceptable justification.	Justify on 2 resources with acceptable justification.	Justify on 3 resources with very significant justification.	Justify on more than 3 resources with very significant justification.



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COMPLEX ENGINEERING ACTIVITIES ATTRIBUTES	Descriptors for Rubrics Design	EA (Some or all)	Level-1 Needs Work	Level-2 Developing	Level-3 Satisfactory	Level-4 Competent	Level-5 Excellent
Level of interactions = interactions between wide & conflicting technical, engineering & other issues	Adapt significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues.	EA2	Associate with 1 level of interaction.	Adapt 1 level of interaction.	Associate with 2 levels of interaction.	Associate with 3 levels of interaction.	Associate with more than 3 levels of interaction.
	Justify the solutions achieved arising from interactions involving wide ranging or conflicting technical, engineering or other issues.		Discuss the solutions based on the 1 level of interaction	Justify the solutions based on 1 level of interaction	Justify the solutions based on 2 levels of interaction	Justify the solutions based on 3 levels of interaction	Justify the solutions based on more than 3 levels of interaction



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COMPLEX ENGINEERING ACTIVITIES ATTRIBUTES	Descriptors for Rubrics Design	EA (Some or all)	Level-1 Needs Work	Level-2 Developing	Level-3 Satisfactory	Level-4 Competent	Level-5 Excellent
Innovation = creative use of engineering principles and research-based knowledge	Advocate creative use of engineering principles and research-based knowledge in novel ways	EA3	Conceptualize only 1 creative principle used.	Conceptualize 1 creative principle used but do not elaborate on the novelty.	Advocate 1 creative principle used and elaborate on the novelty.	Advocate 2 creative principles used and elaborate on the novelty.	Advocate 3 creative principles used and elaborate on the novelty.
	Justify creativity towards the achievement of the novelty (e.g. patent/copyright/etc.)		Justify the 1 creative principle used.	Justify the 1 creative principle used but do not elaborate on research-based knowledge.	Justify the 1 creative principle used and elaborate on research-based knowledge.	Justify the 2 creative principles used and elaborate on research-based knowledge.	Justify the 3 creative principles used and elaborate on research-based knowledge.



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Consequences to society and environment = significant consequences, characterized by difficulty of prediction & mitigation	Organize significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.	EA4	Organize and characterize 1 context	Organize and characterize only 1 difficult context	Organize and characterize 2 difficult contexts	Organize and characterize 3 difficult contexts	Organize and characterize more than 3 difficult contexts
	Exemplify significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.		Justify the consequences	Justify (1) difficulty and its consequences	Justify (2) difficulties and their consequences	Justify (3) difficulties and their consequences	Justify the (>3) difficulties and their consequences



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Familiarity = extend beyond previous experiences using principle-based approaches	Organize resolution beyond previous experiences routinely encountered &	EA5	Organize by applying 1 principle-based approach	Organize by applying 1 principles-based approach beyond previous experience.	Organize by applying 2 principles-based approaches beyond previous experience.	Organize by applying 3 principles-based approaches beyond previous experience.	Organize by applying more than 3 principles based approaches beyond previous experience.
	Exemplify experiences in resolving the engineering activities.		Justify the approach during the resolution.	Justify the approach during resolution beyond previous experience	Justify the approaches (2) during resolution beyond previous experience.	Justify the approaches (3) during resolution beyond previous experience.	Justify the approaches (>3) during resolution beyond previous experience.



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Complex Engineering Activity Rubrics (Form)		
Project Title:	Design and implement an FPGA-based encryption/decryption unit (e.g., AES, RSA, SHA) for secure communication AES: Advanced Encryption Standard, RSA: Rivest-Shamir-Adleman, SHA: Secure Hash Algorithm	
Subject	Digital Systems Design Lab	
Submitted by:		
Names		Registration #

Mapping with Graduate Attributes

Sr. #	Program learning Objective (PLO)	Course learning Outcome (CLO)	Complex Engineering Activities Attributes	Marks		
1	PLO3	CLO3	EA2, EA3			
2.	PLO5	CLO4	EA1, EA5			
3.	PLO9	CLO5	EA4			
4.	PLO10	CLO6	EA5			